Predicting Rally Outcomes and Set Win Probabilities in NCAA Volleyball

Ethan Leap,[1] Sydney Stanton,[1] Rishi Nair,[1] and Chase Meining[1]

[1] *Department of Statistics & Data Science, University of Colorado Boulder*

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This study presented an approach to predict rally and set win probabilities in volleyball by integrating a generalized additive model (GAM) with a Markov chain simulation. Prior studies have used only men's or professional volleyball data, whereas this study focused on women's collegiate volleyball data from the University of Colorado Boulder (CU). Our research quantified patterns previously based solely on intuition to inform evidence-based strategies in collegiate volleyball. We implemented a GAM to predict rally win probabilities, then fed them into our Markov chain simulation to output set win probabilities from CU's perspective. The components of our modeling captured the effects of the serving team and score differential. Our results demonstrated that this integrated approach improves probability accuracy in women's collegiate volleyball compared to past research. The findings underscored the effectiveness of our probability tree visualization in enhancing in-game coaching by providing data-driven insights into win probabilities across different match scores.

1. INTRODUCTION

In this section, we introduce background information, past literature, and our proposed research question.

1.1. Background

Sports analytics has emerged as a more precise and rigorous way to track win probabilities and scoring systems within the sports industry. While win probability models are increasingly prevalent in sports like football, basketball, and baseball, less is known about their behavior in other sports. In particular, analytical approaches could be beneficial during the most decisive moments of a collegiate volleyball match. Statistical modeling can aid in generating effective strategies and forecasting game outcomes. Our focus is on determining live win probabilities throughout a match using six years of women's volleyball data from the University of Colorado Boulder (CU).

Volleyball is a rally-scoring sport, starting with a serve and ending with a point on every play, unlike other sports in which a team must be serving to win the point. In this form of play, point swings are more frequent and can more dramatically impact the result of sets and, therefore, the match. In collegiate volleyball, matches are played to the best-of-five sets, with each set won by the first team to reach 25 points. If a fifth set, which is known as the deciding set, is necessary, it is played to 15 points.

Along with that, a key concept is side-out success, which occurs when the team that receives the serve wins the rally and scores a point. Significant point runs for the serving team can happen when a receiving team's side-out success dips. Therefore, coaches must be aware of how these events affect the probability of winning the set and, by extension, the match.

A single point in a match can be a pivotal moment. This crucial moment can be thought of as a high-leverage score, occurring when a single point has a disproportionately large effect on a team's probability of winning a match. Understanding how win probabilities shift within a set and across the match can help make crucial decisions during the most pivotal moments. By analyzing changes in win probabilities across different scores in sets, this study aims to identify opportunities for more evidence-based decision-making during critical moments in a match.

1.2. Literature Review

Research on real-time win probability models for volleyball has been previously done [1]. Hawkins, Fellingham, and Page (2025) started by computing a baseline of a team's probability of winning a set using pre-set strength metrics. Then they developed a covariate-informed product partition model (PPMx) using in-set performance numbers (hitting percentage, serve efficiency, block percentage, and dig percentage). Finally, the group constructed a Monte Carlo score table to factor in the probability of winning a set based on only the score. They blended these three by weighting the first points of a set entirely on pre-set metrics. At the start of a set, the model gives most of the weight to the team's pre-set ability, based on historical performance. As rallies progress, the weight gradually shifts toward in-set performance, reflecting how the team is currently playing. Around the midpoint of the set (about 16 rallies in), the weighting is roughly 0% pre-set, 90% in-set, and 10% score. In the later stages, as the set nears its conclusion, the focus shifts almost entirely to the score, reflecting how close each team is to winning (about 10% in-set and 90% score). This dynamic weighting system allows the model to adapt its emphasis over time: it starts by relying on historical strength, transitions to in-game performance,

and finishes by prioritizing the score as the key indicator of win probability.

Similar to the paper above, Shoval and Barron (2021) focused on a rotation-based probabilistic model using a Markov chain in which each state represents the serving team's current rotation paired with the opposition's receiving rotation[2]. Unlike the approach above, it allows for each rally-win probability to vary by rotation rather than assuming a single constant rate. This research used data from the Australian National Team and the Israeli League and showed that rotation-specific probabilities can reliably predict set scores within a few points. This work highlights that team success is strongly tied to rotational structure, suggesting that models of win probability must account for these differences, rather than treating all rotations as equal.

While both of the above studies provided valuable frameworks for modeling win probability, they also highlighted the challenge of balancing accuracy with interpretability. To address this, Ntzoufras, Palaskas, and Drikos (2019) developed a Bayesian model specific to volleyball[3]. Their primary response variable is set difference, the difference between sets won and sets lost in a match. They adapted their model to the discrete scoring margins in volleyball. They then implemented a clustering technique called SALSO, developed by Dahl, Johnson, and Müller (2022)[4]. This search algorithm identifies representative partitions from posterior samples of Bayesian clustering models. We implemented similar techniques to identify pressure points using easy-to-interpret categories.

1.3. Research Question

Although prior research has modeled win probabilities in many contexts, it has never been explicitly applied to collegiate volleyball data. Our primary research question is: *How do win probabilities shift as teams progress through a set, from early rallies to late, high-pressure points, and across different sets?*

This research question is important because win probability models are becoming increasingly prominent in sports analytics, yet little is known about their behavior during the most pivotal moments in volleyball competition. Understanding how win probabilities shift at key moments will help coaches make more informed strategic decisions, improve athlete performance under pressure, and enhance competitive outcomes. If this question remains unanswered, teams may continue to rely on intuition rather than data-driven insights during critical moments, potentially missing opportunities to optimize performance and strategy.

We also set out to answer sub-research questions such as the following:

1. *Are there high-leverage scores?*
2. *Are there ways of grouping scores together into discrete categories by their win probabilities?*
3. *How do win probabilities differ between non-deciding sets and deciding sets?*

The next section of the paper discusses our methodologies in data cleaning, formatting, approaches, and modeling validation. In Sec. 3, we discuss the results, including data exploration and summary statistics, assumption and diagnostic testing, model performance, and model validation. In Sec. 4, we discuss the real-world applications of our model and how it can be built upon in the future.

2. METHODS

In this section, the tools, data cleaning and transformation, model architecture, model assumptions, and model validation are discussed.

2.1. Tools

This study was conducted using R 4.4.3 and GitHub for version control to store, share, and collaborate on code. The packages *tidyverse, knitr, tinytex, dplyr, ggplot2, readr, tibble, here, car, visreg, ResourceSelection, randtests, purrr, stringr, mgcv, pROC, viridis*, and *viridisLite* were loaded at the beginning of the workflow and used throughout the code. The *tidyverse* packages *dplyr, ggplot2, readr*, and *tibble* were used for data wrangling and exploratory analysis. The packages *knitr* and *tinytex* were utilized for formatting. The package *car* was used for Box–Tidwell testing. The package *here* was used for consistent file paths throughout the project. The package *visreg* produced a visual for a partial residual check. The packages *randtests, ResourceSelection*, and *pROC* were used for statistical testing and validation. The package *mgcv* was used for generalized additive modeling. Lastly, *viridis* and *viridisLite* were loaded to create a cohesive color scheme for final visualizations.

2.2. Data Cleaning and Transformation

The dataset used is from CU Boulder Athletics, which provided roughly six years of women's volleyball match data, collected internally by coaches. Official NCAA records can confirm all of the information in the dataset we used. The dataset includes 269 628 observations from August 24, 2018, to October 3, 2024. The data source is reputable, as it was recorded internally by the CU volleyball coaching staff. We had no challenges accessing

the data, but others will have to request access to the datasets.

Our raw dataset consisted of touch-level data, meaning every time a player came into contact with the ball. This allowed the dataset to have descriptors such as skill type and evaluations of those skills. However, our project was not concerned with this specific information. Any missing data occurred in variables irrelevant to our analysis, and there were no outliers. Thus, the dataset is complete and consistent with formatting; therefore, no imputation was needed.

During data cleaning, the raw `all_matches.csv` file was imported, normalized, and restructured into rally-level observations suitable for modeling. To ensure data integrity and reliability for subsequent analysis, the data underwent a comprehensive cleaning process. The season variable was standardized to a two-digit format, and all touches were sorted by match, set, and touch index to ensure we followed the in-game sequence. A binary "rally-end" flag was generated whenever the home or visiting teams' score changed from the previous touch. The sum of these flags produced a unique rally ID for each point. These IDs indicate the order of rallies within each set, allowing us to examine the Nth point across all sets, thus ensuring rallies were chronologically ordered within each match and preserving event order.

Missing or empty score entries were filtered out early on. Then the multiple touch-level records were collapsed into single rally-level rows, which summarized key variables such as final scores, serving team, substitutions, and timeouts. Next, a logical helper function was created to standardize Boolean fields, such as timeouts. New variables were derived to represent CU-specific context, including score differentials, serve status, and rally outcomes. At the same time, variables like pre-rally and post-rally scores were computed to capture transitions. Additional transformations identified deciding sets, late-set rallies, and close-score situations to quantify match context and high-pressure situations. Momentum-related features, such as rally run length and previous rally outcome, were added as well for in-game dynamics. Although momentum-related variables were not used in our modeling, they were still explored. Finally, the cleaned rally-level dataset was uploaded to our GitHub repository.

The final dataset, `points_final.csv`, is structured as a long-format, point-by-point volleyball match dataset, with each row representing a single contact. The final cleaned dataset contains 27 580 observations and 30 variables, encompassing 169 matches across seven seasons and 628 unique sets. The dataset contains a mix of categorical, numerical, and Boolean variables, including match and set identifiers, match dates, point indices, score differentials, and binary indicators for serving team, rally winner, and set outcome.

2.3. Assumptions

We begin by assuming that the dataset is accurately recorded and free of errors. Because our data consists exclusively of CU volleyball matches, we treat it as a representative depiction of that team's performance, though not necessarily of collegiate volleyball as a whole. We further assume that rules, scoring structures, and match formats have remained consistent over the period in which the data were collected.

Our analysis also assumes that each rally outcome is conditionally independent of the pre-rally score differential and serving status. This simplifies our modeling, as it does not capture potential momentum effects, psychological influences, or team dynamics that may introduce dependencies across rallies.

From a modeling standpoint, logistic regression assumes that the log-odds of winning a rally are linearly related to the predictors. To address potential nonlinearities, we incorporate a quadratic term for the score differential. The GAM further relaxes this linearity assumption by allowing smooth, data-driven relationships between predictors and the log-odds of a rally win. The Markov chain model, in turn, operates under the Markov property, assuming that the future state of a match depends only on the current score and the serving team, not on the sequence of prior rallies.

Lastly, we assume that external influences such as home-court advantage, officiating decisions, or player fatigue do not meaningfully distort rally outcomes. Our analysis focuses on results from an idealized "perfect game" framework, while recognizing that such real-world factors may still have an influence in practice.

2.4. Model Architecture

Our approach to predicting set win probabilities involves two main steps: first, a GAM predicts individual rallies, and second, a Markov chain model calculates set win probabilities based on the GAM outputs.

Before switching to a GAM, we began with a logistic regression model. The logistic regression model's framework was constructed to estimate the probability that CU wins a rally, given the pre-rally score differential and the serving team. This model serves as the foundation for a Markov chain model of in-game win probabilities based on the previously stated variables. To predict rally outcomes, we created a logistic regression model in which the log-odds of CU winning a rally are modeled as a function of the pre-rally score differential and a binary indicator for serving. The general form of our model is:

$$\text{logit}(p) = \beta_0 + \beta_1 d_{\text{pre}} + \beta_2 d_{\text{pre}}^2 + \beta_3 \,\text{CU}_{\text{serv}}.$$

In this equation, p represents the probability that CU

wins the rally, and β_0 represents the baseline log-odds of CU winning a rally when the score is tied. The next coefficient, β_1, captures the linear effect of the pre-rally score differential, including how much the log-odds of winning change for each additional point to the leading team. Including a quadratic term, $\beta_2 d_{\text{pre}}^2$, allows for this relationship to account for nonlinear effects. This permits the score differential to increase or decrease by varying magnitudes of lead or deficit. Finally, β_3 measures the change in log-odds associated with CU serving compared to when the opponent is serving.

This model generated predicted probabilities across score differentials ranging from a deficit of 24 to a 24-point lead for both serving and receiving situations. These probabilities describe how the likelihood of a winning rally evolves as the match score changes and serve possession alternates.

We also fit a different rally-level probability model using a GAM to compare with the logistic regression model. This approach provides more flexibility, instead of assuming a quadratic relationship in the data. Formally, the GAM can be written as:

$$\text{logit}(p) \;=\; \beta_0 + f(d_{\text{pre}}) + \beta_1 \cdot \mathbb{I}\{\text{CU serve}\},$$

where $f(\cdot)$ is a smooth spline function estimated from the data and $\mathbb{I}\{\cdot\}$ is an indicator function for serving. We expect both models to be similar and plan to use the better-performing method in our final model.

Once these rally-level probabilities were generated, we used them to create a Markov chain model to simulate the progression of a volleyball match. In this chain, a state can be defined as (a, b, s), where a is CU points, b is opponent points, and $s \in \{\text{CU}, \text{Opp}\}$ is the serving team. The chain absorbs if $a \geq T$ and $a - b \geq 2$ (CU wins) or $b \geq T$ and $b - a \geq 2$ (opponent wins), with $T = 25$ (or 15 in the deciding set). Let $W(a, b, s)$ be the probability that CU wins the set from (a, b, s). Then:

$$\begin{aligned} &W(a, b, s) \\ &= p(d, s)\, W(a+1, b, \text{CU}) + [1 - p(d, s)]\, W(a, b+1, \text{Opp}), \end{aligned}$$

where $d = a - b$ (score differential) and s is whether CU is serving. We set $W = 1$ in CU-win absorbing states and $W = 0$ in CU-loss absorbing states.

To avoid extreme probabilities at uncommon score differentials, we applied a clamp at ± 12 score differential. We determined that this was the range of values with sufficient occurrences in the data. We were not confident enough to use the instances outside of this range as predictions. In application, any state outside the range of -12 to 12 uses the transitions of either -12 or 12, instead of a much more extreme value.

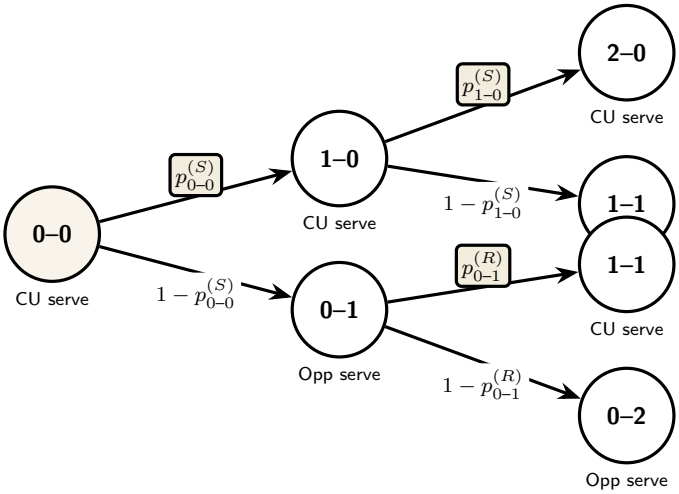

FIG. 1. Example of early CU volleyball rallies based on serving team in a Markov chain.

2.5. Model Assumptions and Model Validation

We assessed the assumptions of our models systematically. For logistic regression, we checked multicollinearity among predictors, verified linearity in the logit for continuous variables, and ensured independence of observations by focusing on rally outcomes under an idealized framework. For the GAM, we evaluated the smooth terms for significance and effective degrees of freedom, assessed model flexibility to avoid under- or over-smoothing, and tested additivity by comparing a model with separate smooths for each level of a binary factor to our original additive model. Finally, for the Markov chain model, we examined lagged rally outcomes to confirm that including previous rallies did not improve predictive performance beyond the current score and serving state, supporting the Markov assumption for rally transitions.

We also have a few assumptions when implementing our clamp into the Markov chain. The first is that the rally-level win probabilities do not differ significantly from the trend at the extreme values. This is a reasonable assumption because these high-score differentials are not only rare but also appear to behave similarly across most states, as expected. The clamp, therefore, represents a deliberate bias toward realism and numerical stability. We prefer to treat rare, unobserved scoring extremes as if they were the largest realistic observed states, rather than allowing the model to extrapolate beyond the domain supported by the data.

Finally, model validation was conducted on the GAM using a five-fold cross-validation procedure at the match level to prevent data leakage between the training and test sets. For each fold, the model was trained on four-fifths of the matches and evaluated on the remaining fifth. Predictive accuracy was assessed using three standard metrics,

which were AUC, log loss, and Brier score. AUC (area under the ROC curve) indicates discriminative ability. Log loss captures probabilistic calibration and penalizes overconfident errors. The Brier score measures mean squared error between predicted and actual outcomes. Across all folds, these metrics provided an estimate of model performance. Finally, a calibration analysis was performed using a plot to assess whether the model's probabilistic predictions aligned with empirical tendencies. Together, these procedures validated the model's generalization, reliability, and interpretability in translating rally-level predictions into set-level win probabilities.

3. RESULTS

We evaluated our proposed Markov chain model and its performance using point-by-point data from women's collegiate volleyball. The model estimates the probability that CU wins a set for every possible score state in a match, starting with calculating the likelihood of every rally while accounting for the serving team and score differential.

3.1. Data Exploration and Summary Statistics

Before fitting the model, we examined the variables to confirm that they were consistent with our assumptions. Rally outcomes were balanced across the dataset, with CU winning about 50.36% of all points, regardless of who was serving. When comparing serving and receiving, CU won about 42.04% of points while serving and about 58.35% while receiving. This is also known as CU's side-out success rate, which supports prior research on a slight advantage for receiving the serve in women's collegiate volleyball [**?**]. The disadvantages of serving include serving errors, the opposing team's ability to neutralize serves with three touches, and the first "turn" to attack.

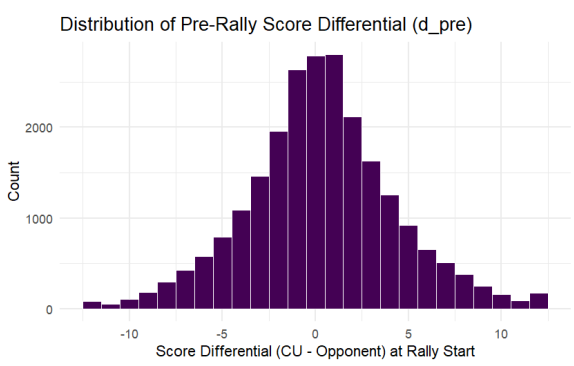

FIG. 2. Distribution of score differential across all rallies.

CU served 50.19% of the rallies, with each team serving roughly an equal amount of the time. The score-differential variable had approximately a symmetric distribution around zero, as seen in Fig. 2, indicating that the dataset captures a balanced mix of leading and trailing situations. This ensures that both the logistic regression and GAM models learn from a full spectrum of competitive states. As the score differential increases, CU's rally-win probability rises smoothly and consistently, supporting the appropriateness of the logistic form and confirming that the GAM effectively captures this nonlinear yet monotonic relationship between lead size and winning likelihood.

3.2. Diagnostics and Assumptions Met

The logistic regression assumptions were tested first. All of the variance inflation factors (1.071186, 1.013266, 1.057551) were close to 1, indicating little to no multicollinearity. Additionally, our dataset has a large enough sample size of 27 580 observations. We then performed the Box–Tidwell test to confirm linearity in the logit form. Only the pre-rally score differential was tested for nonlinearity, as it was the only continuous predictor. The relationship between pre-rally score differential and log-odds was deemed approximately linear, confirming this assumption. Next, we produced a visual for a partial residual check, which showed a roughly smooth, monotonic curve. Lastly, because our analysis focuses on modeling rally outcomes within an idealized "perfect game" framework, we do not account for momentum or fatigue, as previously mentioned, in order to maintain the independence of observations.

For the GAM, we examine the summary output, where the smoothed term is significant at $\alpha = 0.01$ and the effective degrees of freedom are about 1, indicating that the linearity assumption in the link function holds well. A k value of 9 confirms that the model has sufficient flexibility to fit the data but is not underfitting or oversmoothing. In a new GAM fit, we test for additivity by using a separate smoothing factor for the pre-rally score differential in each serving condition, while also including the main effect of CU serving. Our additive model is then compared with the newly created interaction (non-additive) model. The interaction model has a higher AIC (37419.51) than the additive model (37410.47), indicating a worse model fit. The deviance difference (-8.0992) is negative, also suggesting that adding the interaction actually made the model fit slightly worse. Therefore, the simpler additive model is preferred and fits slightly better than the more complex interaction model, thereby satisfying the additivity assumption.

Our Markov chain model was then tested to make sure Markov assumptions held for our rally-level win probability model. The Markov assumption implies that the outcome of the next rally depends only on the current state of the score and the serving team, not on previous rally outcomes. Lag variables were created to represent

the results of three earlier rallies within each match and set. An extended logistic model including the lag variables was then fitted to compare with our Markov model. The inclusion of lagged rally outcomes did not improve the model after accounting for the score differential and serving team. This suggests that past rally outcomes provide no additional predictive power beyond the current state, supporting the Markov assumption in our win probability model.

3.3. Model Performance and Findings

To evaluate the performance of our proposed model, we first compared our logistic regression to our GAM. Both models were built into the Markov chain framework to calculate point-by-point transition probabilities and overall set win probabilities. The logistic regression served as a baseline model, whereas the GAM provided a more flexible model that could capture the nonlinear effects of score difference and serving status. Because of that flexibility, we opted to use the GAM in our final Markov chain. To prevent overfitting at extreme values, we clamped the Markov chain at a score differential of ± 12. For example, if CU leads by 15, we instead use the transition state when CU leads by 12 to avoid instances where there aren't enough data points at a given score differential, since the model would not return a confident prediction. Therefore, Fig. 3 displays all predicted win probabilities in the score differential threshold range compared with empirical values, differentiating by which team is serving. The size of the points on the plot represents the number of observations at each score differential in the dataset.

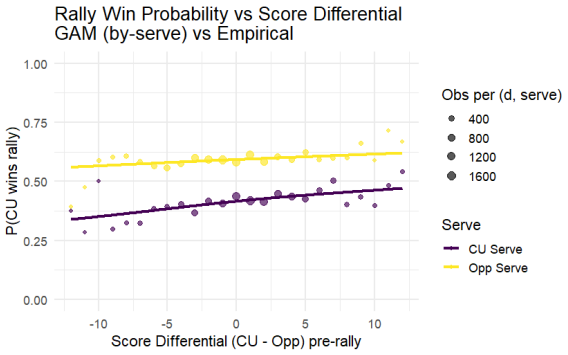

FIG. 3. Predicted versus empirical rally win probabilities across score differentials, separated by serving team. Point size reflects the number of observations.

Across both models, the predicted win probabilities were very similar, showing that the model is stable and that the score differential and serving team are strong predictors of rally outcomes. As expected, the trends in Fig. 3 are similar when CU serves and when they receive, with the caveat that CU's overall rally win probabilities are higher when the opposing team is serving. This idea matches prior research.

Moving into Fig. 4, which displays set win probabilities, we consider set win probabilities for each score state by serving team. To the left is the entire plot of scores from the beginning to the end of a match, while to the right is a zoomed-in portion of the left plot at the deuce band. The deuce band is scores that are around 22–30 points for both teams. These late rallies are highly decisive, making large impacts on set win outcomes.

At the start of the set, CU's chance of winning the set was about 51–54%, depending on who served first. When CU was the team to serve first, the logistic regression model estimated a probability of winning of about 0.510, while the GAM estimated 0.517. When the opponent served, the probabilities were slightly higher (around 0.535–0.542), reflecting the slight advantage in women's collegiate volleyball that comes with receiving.

In late-set situations, the model showed steep shifts in win probability. At a score of 24–23 and CU serve, CU's chance of winning was roughly 0.74, increasing to 0.82 when the opponent served at the same score. When CU trails 23–24, the win probability dropped to 0.19–0.28, showing how one point near the end of a set can completely swing the expected outcome of a set.

In near-deuce situations—scores tied late in a set, such as 23–23—both models showed win probabilities close to 50%, meaning the outcome could go either way. As soon as one team gained a slight two-point advantage (e.g., 24–22), the predicted chance of winning the set rose sharply to around 90%, reflecting how decisive late rallies become in determining win probability.

Overall, the GAM's predictions differed by less than one percentage point from the logistic model's predictions, confirming that both methods capture the same underlying trends. The logistic model provides a simpler interpretation, while the GAM produces smoother transitions near close scores. Together, they accurately show how serving and score progression shape live set win probabilities. Figure 5 shows an example of a set win probability tree near the end of the set. A probability tree like this can be made more extensively for any other states, but for our purposes, we choose to hone in on one late set win probability tree.

3.4. Model Validation

To validate our model and prevent overfitting in the GAM, a five-fold cross-validation procedure was used to assess the accuracy of rally-level predictions. The dataset was split into five approximately equal groups of matches. For each fold, four groups were used to train the GAM, and the remaining group was used for testing. This process was repeated five times, so that each match served as a test case once. For each test fold, three performance

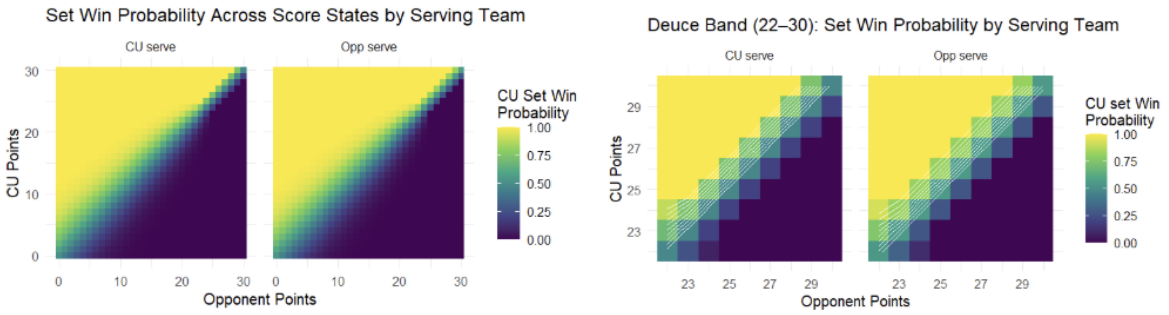

FIG. 4. Heatmaps of set win probabilities across score states by serving team. The right panel zooms into the deuce band (approximately 22–30 points).

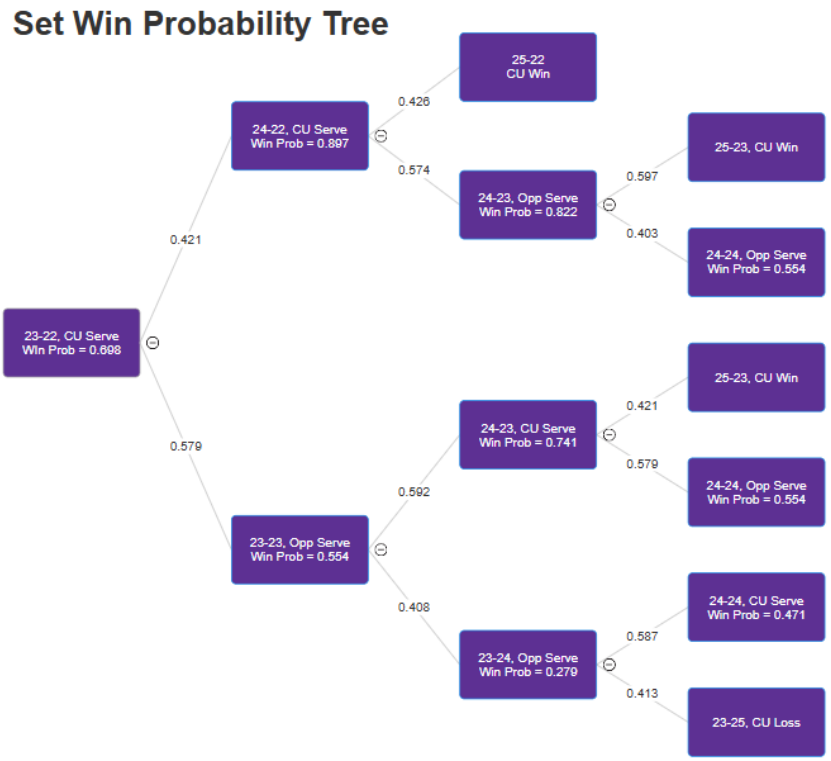

FIG. 5. Win probability tree of close scores at the end of a set.

metrics were computed: AUC, log loss, and the Brier score. The AUC averaged about 0.59, indicating the model performed modestly better than random guessing (AUC = 0.50) and suggesting slight discriminative power. The Brier score (0.243) reflects good calibration for a binary outcome with roughly even win/loss probabilities. The log loss (0.678) suggests the model's probabilistic predictions are well-behaved, avoiding overconfidence in incorrect outcomes. Together, these results demonstrate that while rally outcomes are inherently stochastic, the GAM captures meaningful structure in the data.

TABLE I. Five-fold cross-validation results for the GAM model (AUC, log loss, Brier score).

Fold	AUC	Log Loss	Brier Score
1	0.596 680	0.677 838	0.242 398
2	0.595 272	0.678 262	0.242 602
3	0.598 056	0.677 499	0.242 229
4	0.584 872	0.680 489	0.243 700
5	0.591 468	0.678 094	0.242 511

We then split predicted rally probabilities into ten bins based on their quantiles to create a rally-level calibration table. This table shows how well predicted probabilities

match observed outcomes across the range of predictions, which was then used to develop the rally-level calibration plot displayed in Fig. 6.

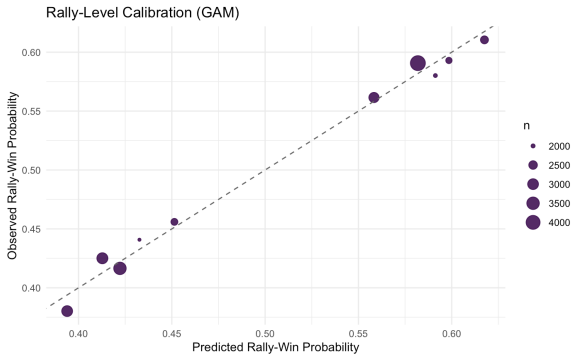

FIG. 6. Rally-level calibration plot comparing predicted and observed rally-win probabilities across ten bins. Point size reflects the number of rallies in each bin.

Also, by simulating 10 000 sets of data, it is seen that the simulated average (54.84%) of the sets won is close to the historical average (53.59%) of sets won. This further indicates that the Markov chain is producing realistic set outcomes. Thus, we conclude that the model is well-calibrated at the set level. To further evaluate the accuracy of our Markov chain model, we calculated a set of weighted Brier scores and set-weighted log loss for all states, and did the same for late set scores (20–20 and higher). Across all states, this evaluation produced a Brier score of 0.3393 and a log loss of 1.1851. These metrics are perfectly reasonable for a model that uses only the score differential and the serving team as predictors. However, we are less interested in predicting early set scores, such as 5–5, so we tested our model only on later set scores. From 20–20 and beyond, we produced a Brier score of 0.3308 and a log loss of 0.9969. These points are far more critical and definitive, so our ability to more accurately model swings in this region is helpful. These results demonstrate that our Markov-chain-driven set-win probability model is well-calibrated and numerically stable.

3.5. Additional Findings

We also used our model to address the sub-research questions.

1. Are there high-leverage scores? In volleyball, a high-leverage score refers to a game situation in which the outcome of the current point has a disproportionately large impact on the team's overall probability of winning the set. Intuitively, these occur in close, late-game situations, such as 24–23, where winning or losing the rally can significantly shift set-win probabilities. To quantify this, we computed a rally-level leverage metric using our Markov chain. For each rally, we evaluated the difference in CU's set-win likelihood before and after the rally, taking the absolute value to capture the magnitude of impact regardless of direction. Larger values correspond to rallies that meaningfully change the set outcome.

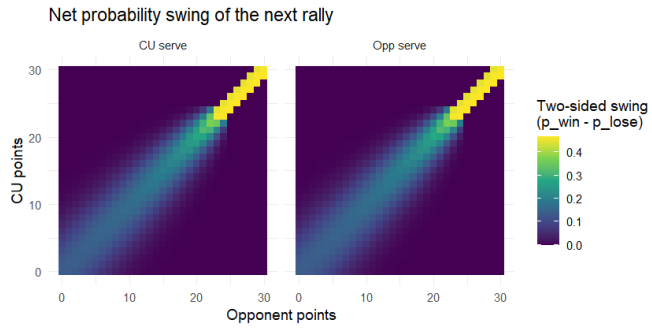

FIG. 7. Next-rally net probability swing heatmap across score states.

Visualizing these leverage scores through changes in set win probability in Fig. 7 reveals that most high-leverage rallies occur when both teams are near the end of the game, confirming the intuition that late, close rallies are critical. In contrast, rallies early in a set have little effect, resulting in minimal leverage scores. This approach provides a quantitative measure of rally importance, as we observe a net probability swing of up to 0.5 in favor of or against CU during late rallies in close sets.

2. Are there ways of grouping scores together into discrete categories by their significance? There are ways to group scores into discrete categories based on similar win-probability swings. The goal was to determine whether score states naturally form distinct groups based on the magnitude of the rally's impact on set outcomes. Figure 8 presents these results by organizing score states into five probability tiers, with each tier representing a different "zone" of the next rally's importance. This would allow for more real-time, actionable choices for coaches, rather than having to scan to an exact state in Fig. 7.

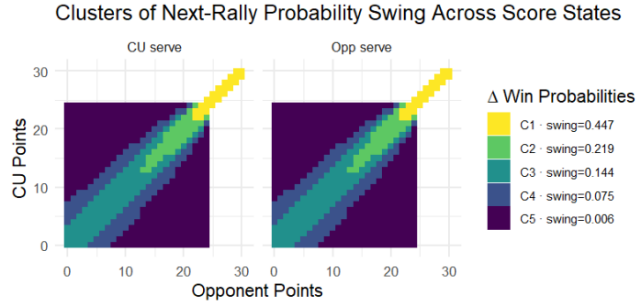

FIG. 8. Heatmap of probability swings grouped into five probability tiers.

The clusters reflect three broad areas of leverage. The

lowest tiers occur early in the set or in extreme-score situations, where rallies are effectively low-pressure events that barely alter the set's trajectory. Intermediate tiers emerge in competitive mid-set states, where rallies begin to shift win probability but remain far from decisive. Finally, the highest tier corresponds to the true late-set "deuce band," where a single rally can produce a significant swing in expected set outcome and meaningfully redefine the likelihood of winning. This clustering method provides a more practical lens for interpreting leverage than relying on raw heatmap gradients.

3. How do win probabilities differ between non-deciding sets and deciding sets? Win probabilities between non-deciding sets and deciding sets do differ. When looking at the distribution of CU points at the set end, we saw that non-deciding sets have more sets and a higher point distribution.

Next, we implemented a standard two-sample t-test using Welch's method, comparing the average CU end score for deciding versus non-deciding sets. The t-statistic was 6.9873, a significant statistic, suggesting a noticeable difference between groups, along with an extremely small p-value $\approx 3 \times 10^{-12}$, indicating the difference in means is statistically significant. Non-deciding set rallies averaged ~ 11.69 points, while deciding set rallies averaged ~ 11.02 points. On average, CU scores slightly fewer points in deciding sets than in non-deciding sets, and this difference is highly significant. The number of rallies per set type differed significantly, so we balanced the sample sizes and re-ran the two-sample t-test. The difference remains substantial even after balancing sample sizes, with a new t-statistic of 6.2343. This suggests that rallies in deciding sets tend to be slightly lower-scoring for CU, which may reflect increased pressure or tighter competition in high-stakes situations.

4. DISCUSSION

In this last section, the domain significance, limitations of our research, future recommendations, and concluding thoughts are discussed.

4.1. Domain Significance

The results of this study can be used to identify how set win probabilities shift over time, as well as to highlight the most critical rallies in a set. These results can be used in real time by coaches and players. Coaches can intervene in several ways, such as calling timeouts, making substitutions, or altering current rotations based on our probability tree and clustered heatmap. Using the clustered heatmap above, coaches could also quickly convey the rally's magnitude to their players. This stresses the importance of a rally, such as a low-risk or must-win rally.

4.2. Limitations

The biggest limitation of this study is that all rally-level data were sourced solely from the CU women's volleyball team. The models we developed learn probability structures from CU's historical performance. The resulting rally-level and set-level win probabilities are team-specific from the perspective of CU. As a result, these probability estimates should not be interpreted as a generalizable framework that other collegiate programs can rely on. Teams with different offensive systems, serving patterns, defensive schemes, or roster compositions could end up with different rally win tendencies. Therefore, the exact set win probabilities reported earlier should not be deployed for predictive use outside the CU context.

However, while the numerical outputs of our models are not transferable, the methodology behind the models can be applied to any team. The data cleaning procedures, the use of a logistic model or GAM approach, and the Markov chain structure can be applied to any team with sufficiently detailed point-by-point data. Ethically, our findings should be used carefully to avoid overstating predictive certainty in such high-pressure environments. Thus, our findings should be used as an additional tool and not an absolute measure. The model captures only statistical tendencies, not psychological factors, injuries, or coaching decisions. Deploying it without acknowledging these constraints could mislead stakeholders.

4.3. Recommendations

Future work could apply our modeling framework to additional teams to examine how rally-level dynamics and set win probabilities vary across programs. Comparing multiple rosters would help determine whether CU's probability structure is typical or whether other teams exhibit distinct patterns in their score-and-serve interactions. Similarly, incorporating shared data between CU and specific opponents would allow matchup-based probability estimation, revealing how the statistical characteristics of two teams combine to shape expected set outcomes.

Our model could also be adapted to produce rally-by-rally set-level win-probability graphs, allowing researchers or coaching staff to visualize how the likelihood of winning evolves throughout a set. This line-chart representation would offer a continuous picture of probability movement and provide clearer insight into the precise score states where leverage peaks.

Finally, the current set-level Markov structure could be expanded to match-level forecasting by embedding the best-of-five format into the simulation process. This

extension would enable the estimation of match-win probabilities, comeback likelihoods, and key transition points across sets, offering a complementary view of how individual rallies shape long-term outcomes.

4.4. Conclusions

Ultimately, our study was set out to understand how win probabilities shift throughout a women's collegiate volleyball set, and we found that they are largely driven by two main factors: score differential and serving team. We did this by using a GAM for rally-win-level probabilities and a Markov chain for set-win probabilities. Throughout the score states, we showed that rallies late in the set produced steep swings in set probabilities. We formally confirmed the existence of these high-leverage rally scores in the deuce band, demonstrating that a single rally could change CU's set win probability by 30–40 percentage points. In contrast, earlier states had much smaller leverage.

By clustering states into tiers of probability swing, we also demonstrated that score states can be grouped into a handful of meaningful rally importance zones, rather than being treated individually for greater and quicker interpretability. Overall, our GAM-driven Markov chain provides a coherent, CU-specific map of how win probabilities evolve within a set, where the same framework can be applied to other teams to support more data-driven coaching and in-game decision making in the future.

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